

ECE 2K7 MINI PROJECT – ELECTRONIC DICE WITH LMC555 AND CD74H4017

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Introduction

The 555 timer, designed in 1971 by Swiss engineer Hans Camenzind for Signetics, is one of the most widely used integrated circuits. Its simple and reliable design has ensured continued use in everything from hobby electronics to industrial systems.

The IC can be configured in multiple modes, but this project will focus on the two most common:

- Monostable Mode: Also known as a "one-shot" multivibrator, this mode has one stable state. When a trigger signal is applied, the output switches to its unstable state for a precise duration, set by an external resistor and capacitor, before returning to its stable state.
- Astable Mode: Also known as a "free-running" oscillator, this mode has no stable states. It continuously self-triggers, producing a rectangular wave at the output. The frequency and duty cycle of this wave are controlled by an RC circuit, typically involving two resistors and one capacitor.

The main objective of this project is to experiment and analyze the modes of operations of the 555 IC and use it with another IC (CD4017) to create a digital dice that is random to the human eye.

Theory

The 555 IC is made up of 4 components – an internal voltage divider, consisting of three $5k\Omega$ resistors (This is where the chip gets its name from), two op amps set up as comparators, a flip flop and a transistor.

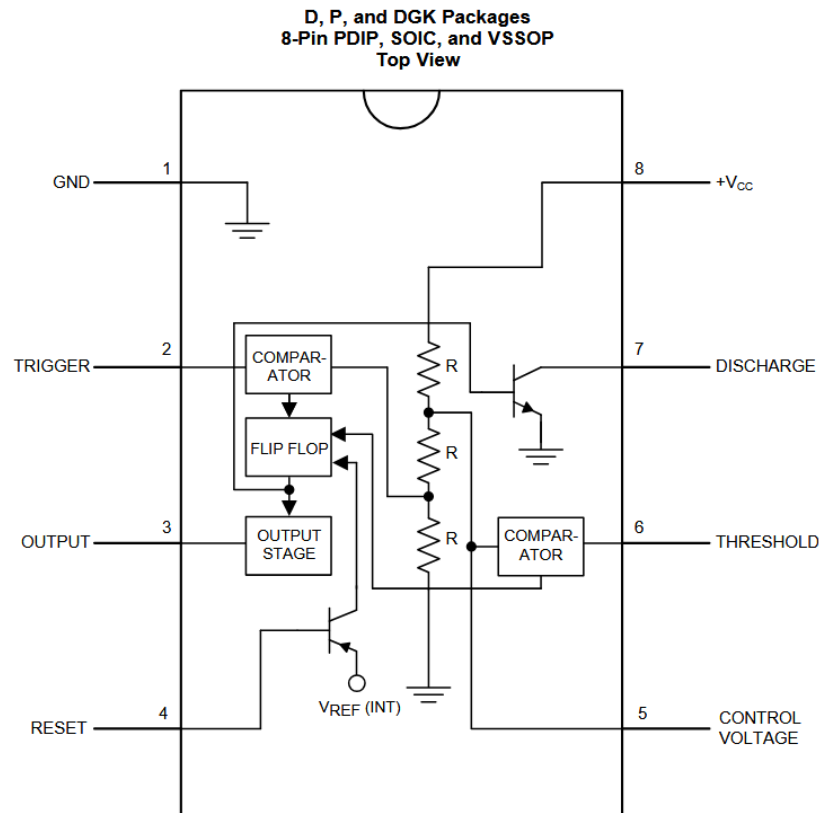


Figure 1: Internal Block diagram of 555

- Internal voltage divider: This is a series of three 5 kΩ resistors between V+ (pin 8) and GND (pin 1). The divider creates two stable internal voltage references $2/3V$ and $1/3V$
- Comparators: Two op amps that are configured as comparators that use the internal reference voltages mentioned above
 - Threshold comparator: This comparator compares the voltage at the Threshold pin (Pin 6) with the $2/3V$ reference. If the voltage at pin 6 is greater than $2/3V$, its output goes high, effectively resetting the circuit
 - Trigger comparator: This comparator compares the voltage at the Trigger pin (Pin 2) with the $1/3V$ reference. If the voltage at pin 2 is greater than $1/3V$, its output goes high, “setting” the circuit.
- S-R Flip-Flop: This is the core digital memory element of the chip. It is "SET" (Q output goes LOW) by the Trigger comparator and "RESET" (Q output goes HIGH) by the Threshold

comparator. The flip-flop's Q output (which is inverted before reaching Pin 3) controls the main output and the discharge transistor.

- Output Stage: An output driver (inverter and amplifier) inverts the S-R flip-flop's state and provides a high-current push-pull output at the Output pin (Pin 3). It can source or sink up to 200 mA.
 - When the flip-flop is SET, the main output (Pin 3) goes HIGH.
 - When the flip-flop is RESET, the main output (Pin 3) goes LOW.
- Discharge Transistor: This NPN transistor is connected to the Discharge pin (Pin 7). Its base is connected to the flip-flop's Q output.
 - When the output (Pin 3) is HIGH, the transistor is OFF, and Pin 7 is "floating," allowing an external capacitor to charge.
 - When the output (Pin 3) is LOW, the transistor is ON, connecting Pin 7 directly to GND and allowing the capacitor to discharge.
- Control Voltage (Pin 5): This pin provides direct access to the $\frac{2}{3} V$ reference voltage. It is typically connected to ground via a $0.01 \mu F$ decoupling capacitor to stabilize the internal reference voltage.
- Reset (Pin 4): This pin when pulled LOW, resets the flip-flop, forcing the Output (Pin 3) LOW and turning ON the discharge transistor (Pin 7). For normal operation, this pin is tied HIGH (to V) to prevent accidental resets.

Monostable mode of operation

In monostable mode (also called “one shot” mode), the 555 creates a pulse of a fixed duration in response of an input trigger

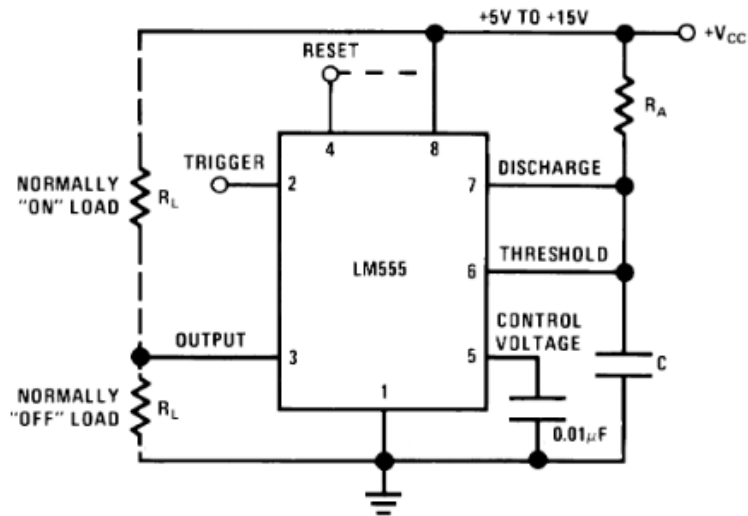


Figure 2: 555 wired up in monostable mode

Operation

- Stable state: The circuit is stable with its output at low when the internal flip flop is low, which means that the discharge transistor (pin 7) is on, the trigger pin (connected to a push button in our experimental circuit) is at HIGH by the pull up resistor.
- Triggering: The circuit is triggered by pulling the trigger pin LOW (putting it below $1/3V$). This fires the trigger comparator, which SETS the flip-flop.
- Unstable State: The flip-flop's state change does two things simultaneously: the Output (Pin 3) goes HIGH, and the discharge transistor (Pin 7) turns OFF. With Pin 7 no longer grounded, the capacitor (C) is free to charge through the external resistor (R_A).
- Return to Stable: The capacitor starts charging to V . As soon as the capacitor voltage (connected to pin 6) reaches the $2/3V$ threshold, the threshold comparator flips, resetting the flip-flop. This returns the circuit to its stable state: the Output (Pin 3) goes LOW, and the discharge transistor (Pin 7) turns ON, discharging the capacitor back to 0V.

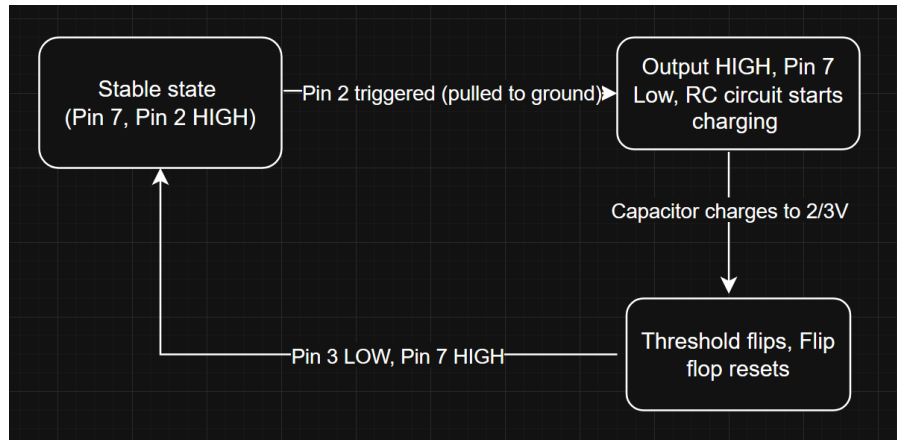


Figure 4: Flow chart of monostable operation

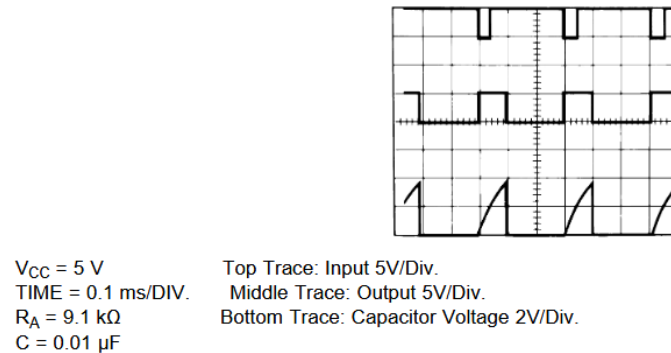


Figure 5: Output signals of monostable configuration

Since the time of the pulse is dependent on the RC circuit, the pulse width is the time it takes for the capacitor to reach 2/3V. Upon further calculations, we get the pulse width time as $T = 1.1 * RC$

Astable mode

In astable, or "free-running," mode, the 555 timer has no stable states and functions as an oscillator, producing a continuous rectangular wave.

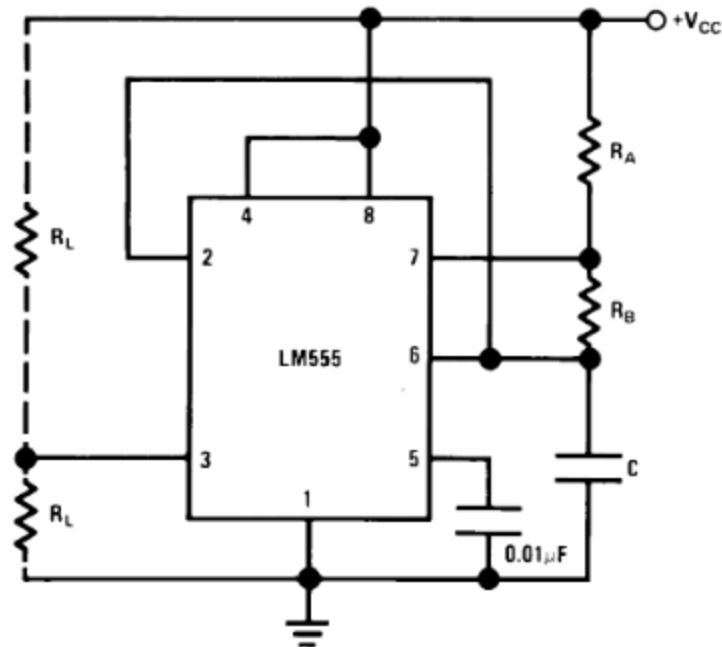


Figure 3: 555 wired up in astable mode

Operation:

This mode is achieved by connecting the Trigger pin (Pin 2) and Threshold pin (Pin 6) together. This allows the capacitor's voltage to trip both comparators, creating a self-sustaining loop.

Charging Phase (Output HIGH): The capacitor voltage starts below $1/3 V$. The Trigger comparator (Pin 2) fires, SETTING the flip-flop. The Output (Pin 3) goes HIGH, and the discharge transistor (Pin 7) turns OFF. The capacitor (C) now charges through both external resistors, R_A and R_B .

Discharging Phase (Output LOW): The capacitor continues to charge until its voltage (monitored by Pins 6 and 2) reaches $2/3 V$. The Threshold comparator (Pin 6) fires, RESETTING the flip-flop. The Output (Pin 3) goes LOW, and the discharge transistor (Pin 7) turns ON. The capacitor now discharges only through resistor R_B into Pin 7.

The capacitor discharges until its voltage drops back down to $1/3 V$. At this point, the Trigger comparator (Pin 2) fires again, SETTING the flip-flop, and the entire cycle repeats.

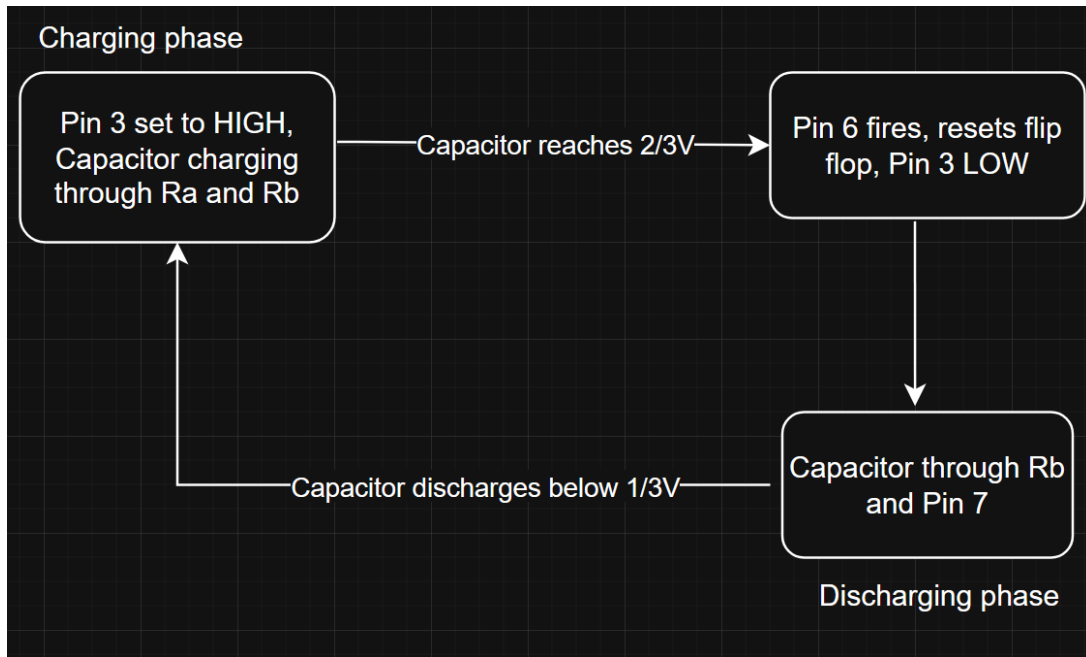


Figure 7: Flowchart of astable operation

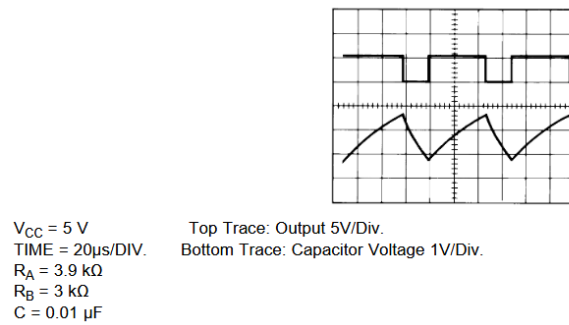


Figure 8: Output signals of astable operation

Key Equations:

The timing is derived from the RC charging/discharging equations between $1/3V$ and $2/3V$.

- Charge Time (Output HIGH): $t_1 = \ln 2 * (R_1 + R_2) * C \approx 0.693 * (R_1 + R_2) * C$
- Discharge Time (Output LOW): $t_2 = \ln 2 * R_2 * C \approx 0.693 * R_2 * C$
- Total Period (T): $T = t_1 + t_2$
- Frequency (f): $1/T$

- Duty Cycle (D): $\frac{t_1}{T} = \frac{R_1 + R_2}{R_1 + 2R_2}$

Experimental procedure

All experiments were conducted with the DC 5V power supply provided in the kit

Monostable mode

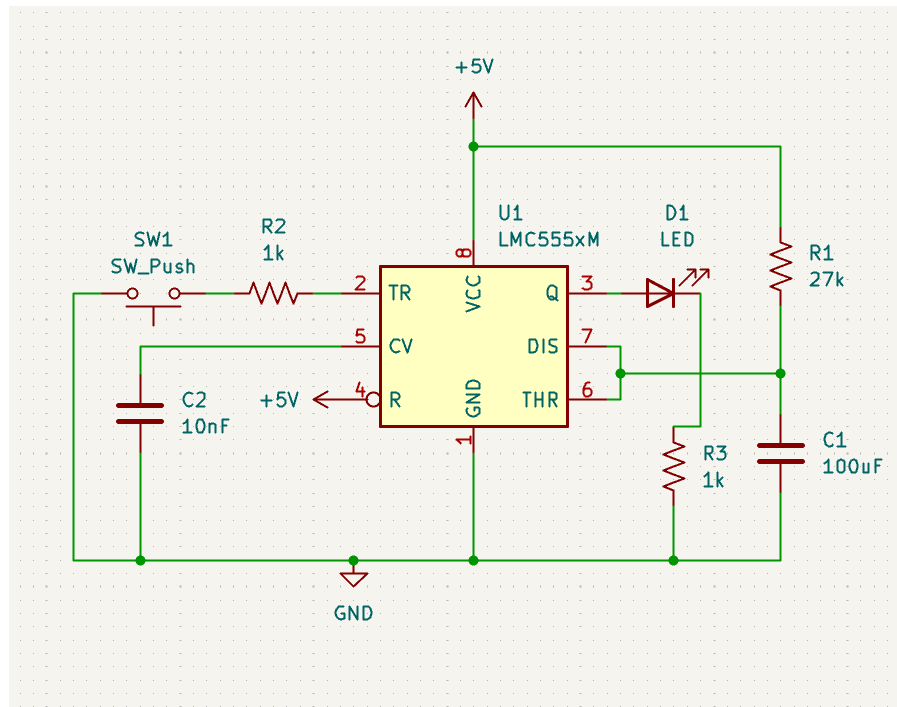


Figure 9: Schematic for monostable mode

Parameter Calculations: The goal is to achieve a high output pulse of 3 seconds.

- Formula: $T = 1.1 * R * C$
- Target: = 3 s.
- Capacitor chosen $C = \mu\text{F}$.
- Solving for R we get $R \approx 27\text{K}\Omega$.
- Select closest resistor value from kit: $22\text{k}\Omega + 4.7\text{k}\Omega + 1\text{k}\Omega$.
- Pulse width with kit resistor values = 3.047S

Circuit setup

- Connect power supply and make power rails
- Short pin 4 and 8 and connect to VCC
- Pin 5 connected to ground with a $0.01\mu\text{F}$ capacitor for decoupling
- Trigger (Pin 2) is connected to ground with a $1\text{k}\Omega$ resistor to debounce the switch
- Output (Pin 3) is connected to an LED with a $1\text{k}\Omega$ resistor to limit the current going through the led
- 5V DC power applied with oscilloscope probe connected at output LED and switch
- Push button pressed to trigger unstable state

Astable mode

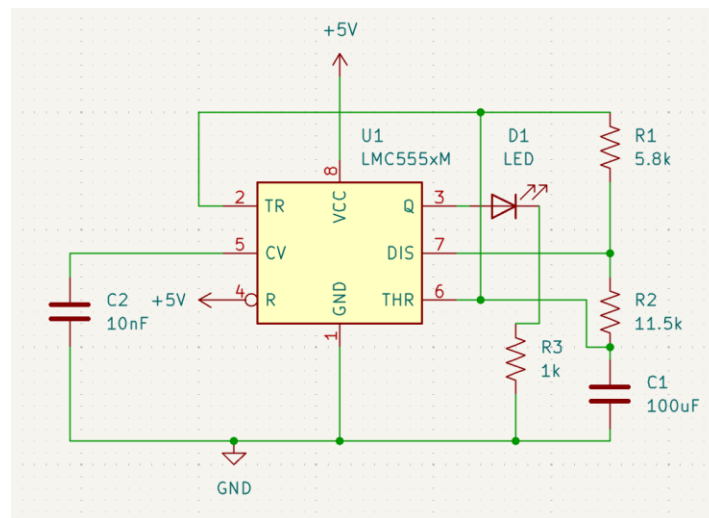


Figure 10: Schematic for Astable mode with 60% Duty cycle over a 2s period

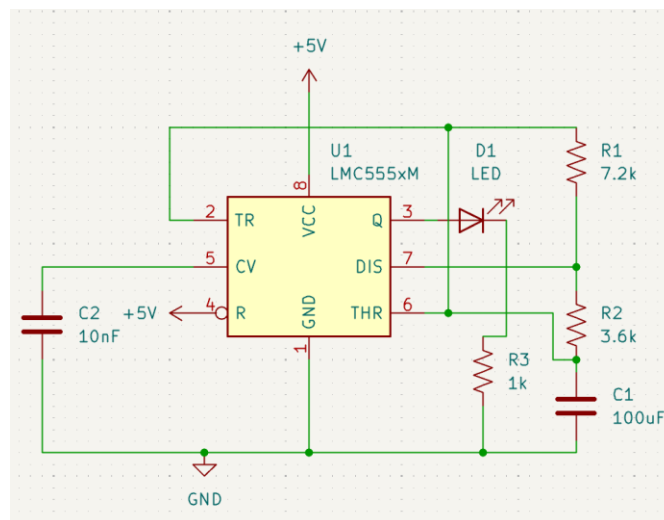


Figure 11: Schematic for Astable mode with 75% Duty cycle over a 1s period

Parameter calculations

- Test 1: 60% duty cycle, 2s period
 - o $T = 2s, t_1 = 1.2s, t_2 = 0.8s$
 - o Using $C = 100\mu F$
 - o $t_2 = \ln 2 * R_2 * C ; R_2 = \frac{0.8}{0.693 * 10^{-4}} \approx 11.5k\Omega$
 - o $t_1 = \ln 2 * (R_1 + R_2) * C ; R_1 = \frac{1.2}{0.693 * 10^{-4}} - 11.5 * 10^3 \approx 5.8k\Omega$
 - o From resistors provided in kit: $R_1 = 5.6k\Omega + 220\Omega, R_2 = 10k\Omega + 1.5K\Omega$
 - o T estimated: $t_1 = 1.24s, t_2 = 0.83$
- Test 1: 75% duty cycle, 1s period
 - o $T = 1s, t_1 = 0.75s, t_2 = 0.25s$
 - o Using $C = 100\mu F$
 - o $t_2 = \ln 2 * R_2 * C ; R_2 = \frac{0.25}{0.693 * 10^{-4}} \approx 3.6k\Omega$
 - o $t_1 = \ln 2 * (R_1 + R_2) * C ; R_1 = \frac{0.75}{0.693 * 10^{-4}} - 3.6 * 10^3 \approx 7.2k\Omega$
 - o From resistors provided in kit: $R_1 = 5.6k\Omega + 1.5k\Omega, R_2 = 3.3k\Omega + 330\Omega$
 - o T estimated: $t_1 = 0.743s, t_2 = 0.251s$

Step-by-step circuit setup and testing:

- The astable circuit was constructed according to the schematic.
- Pin 8 (VCC) and Pin 4 (Reset) were connected to 5V. Pin 1 (GND) to ground and Pin 5 (Control) to ground via a $0.01\mu F$ capacitor.
- Pin 2 (Trigger) and Pin 6 (Threshold) were connected.
- An LED with a $1k\Omega$ resistor was connected to the Output (Pin 3).
- Oscilloscope probe connected to Pin 3 (Output).
- R1, R2 and C connected
- Power connected
- Measure parameters of the resulting waveform.
- Swap out R1,R2 and C with test 2
- Repeat above 2 steps

Adding a potentiometer

By replacing either R1 or R2 with a potentiometer, we can adjust the charge and discharge times of the 555 timer to as much or as little as we want, however, it becomes difficult to calculate the exact time periods for every single resistance possible from the 10k potentiometer

Real life applications of 555 timer

There are endless uses of the 555 timer, however for this project, I have chosen to make an electronic dice with a 555 and a CD4017 Johnson Decade counter, with the 555 providing a pulse and the counter being used to count from 1 to 6

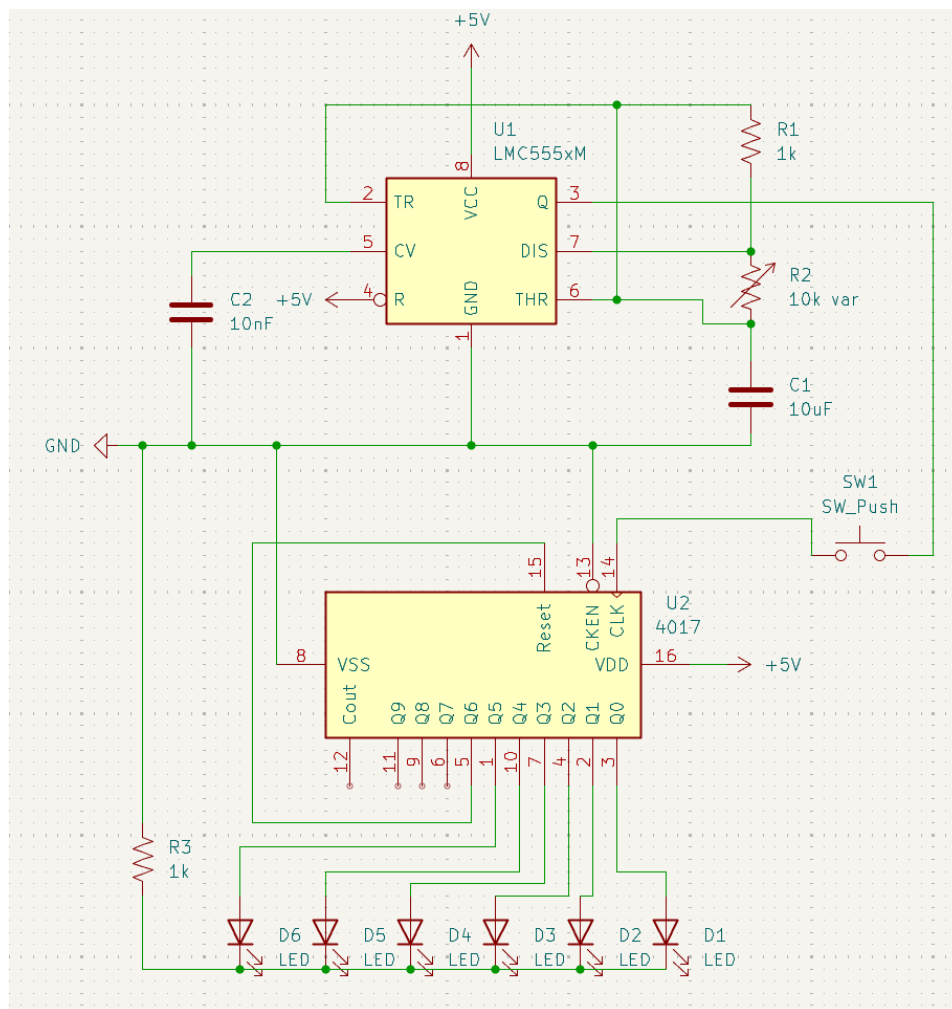


Figure 12: Schematic for Electronic dice with 555 and 4017

Circuit explanation:

- Pulse generation: The 555 timer is configured in astable mode, but with resistances and capacitance low to give it a very high frequency ($\approx 100\text{Hz}$, which is indistinguishable to the human eye). The output of the 555 is connected to the clock pin of the 4017, which advances to the next output.
- Counting: The CD4017 is a counter that switches its output from 1 to 10 when its clock signal hits a peak. To limit the dice to 6 numbers, the 7th output (Pin 6) is shorted to the reset pin (Pin 15). This makes it so that when the counter tried to advance after the 6th output, it automatically resets, going back to the first LED.
- User control: In between the connection to the output pin of the 555 and the clock pin of the 4017 is a push button to be used to “start” and “stop” the dice
- LEDs: 5 basic green LEDs are used, with the 6th LED being red to differentiate between the ends

Results

Monostable mode

Theoretical, simulation and experimental values

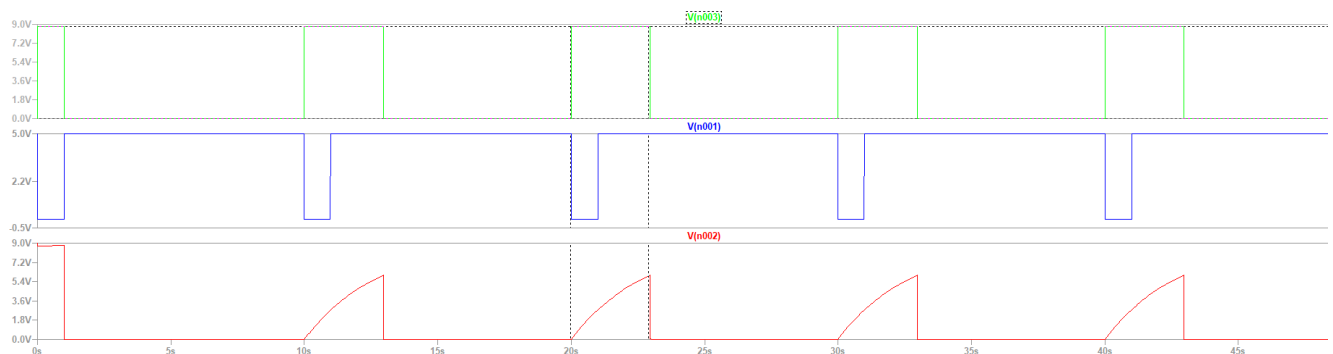


Figure 13: LTspice graph results for monostable simulation

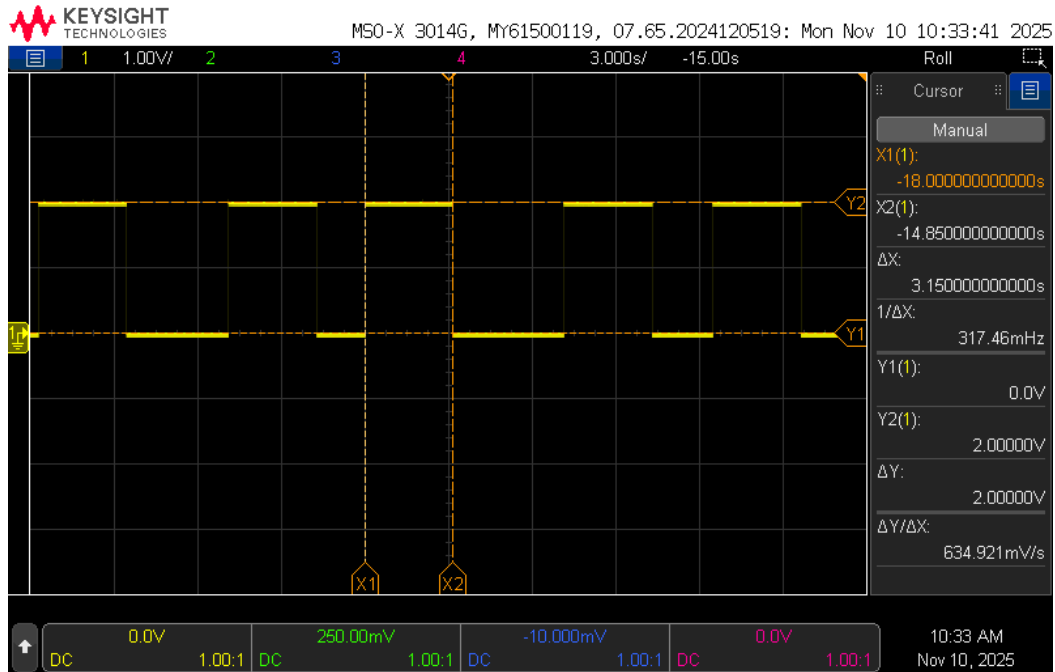


Figure 14: Oscilloscope output for monostable mode

	Pulse width(s)
Theoretical	3.047
Simulated	2.92
Experimental	3.15

Figure 15: Table comparing pulse widths of theoretical, simulated and experimental circuits

Astable mode

Case 1



Figure 16: LTspice graph results for astable simulation in case 1, showing Vout and Vc

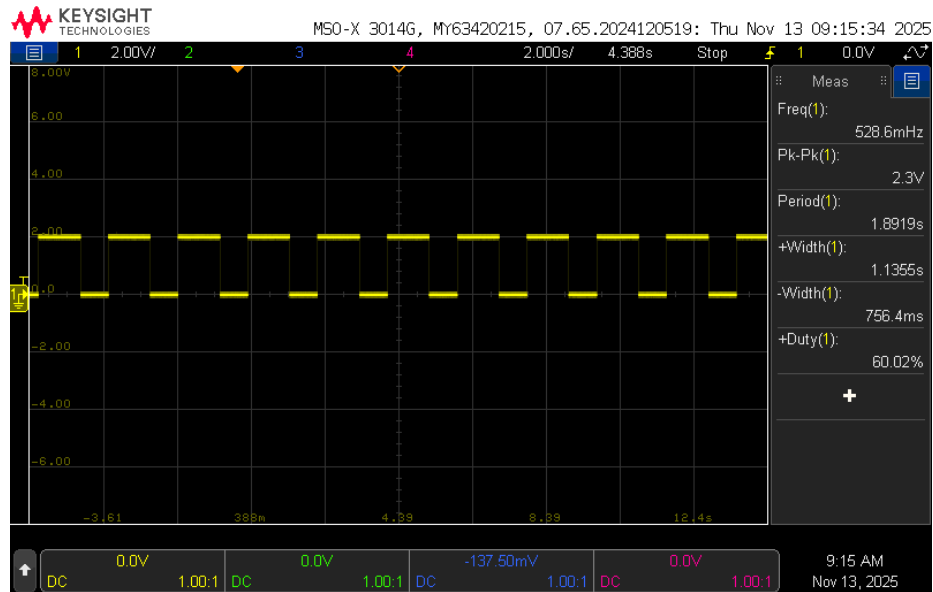


Figure 17: Oscilloscope results for astable simulation in case 1

	T1(s)	T2(s)	T(s)
Theoretical	1.24	0.83	2.07
Simulated	1.206	0.84	2.044
Experimental	1.13	0.756	1.89

Figure 18: Table comparing pulse widths of theoretical, simulated and experimental circuits

Case 2

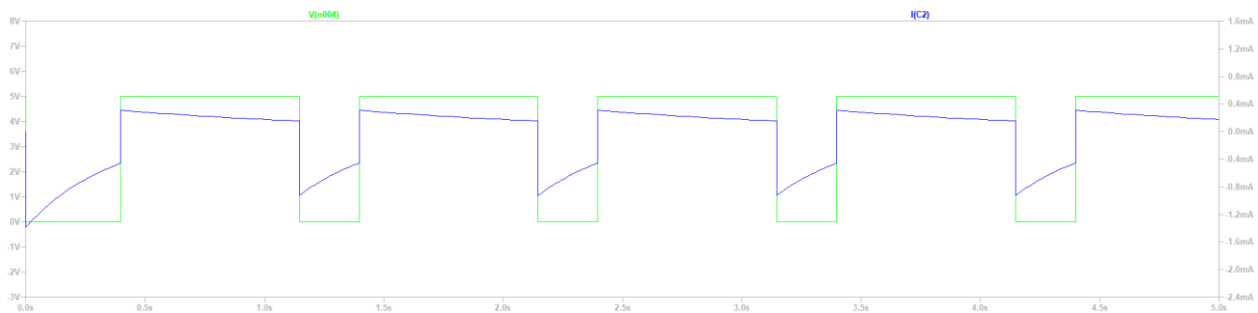


Figure 19: LTspice graph results for astable simulation in case 2, showing V_{out} and V_c

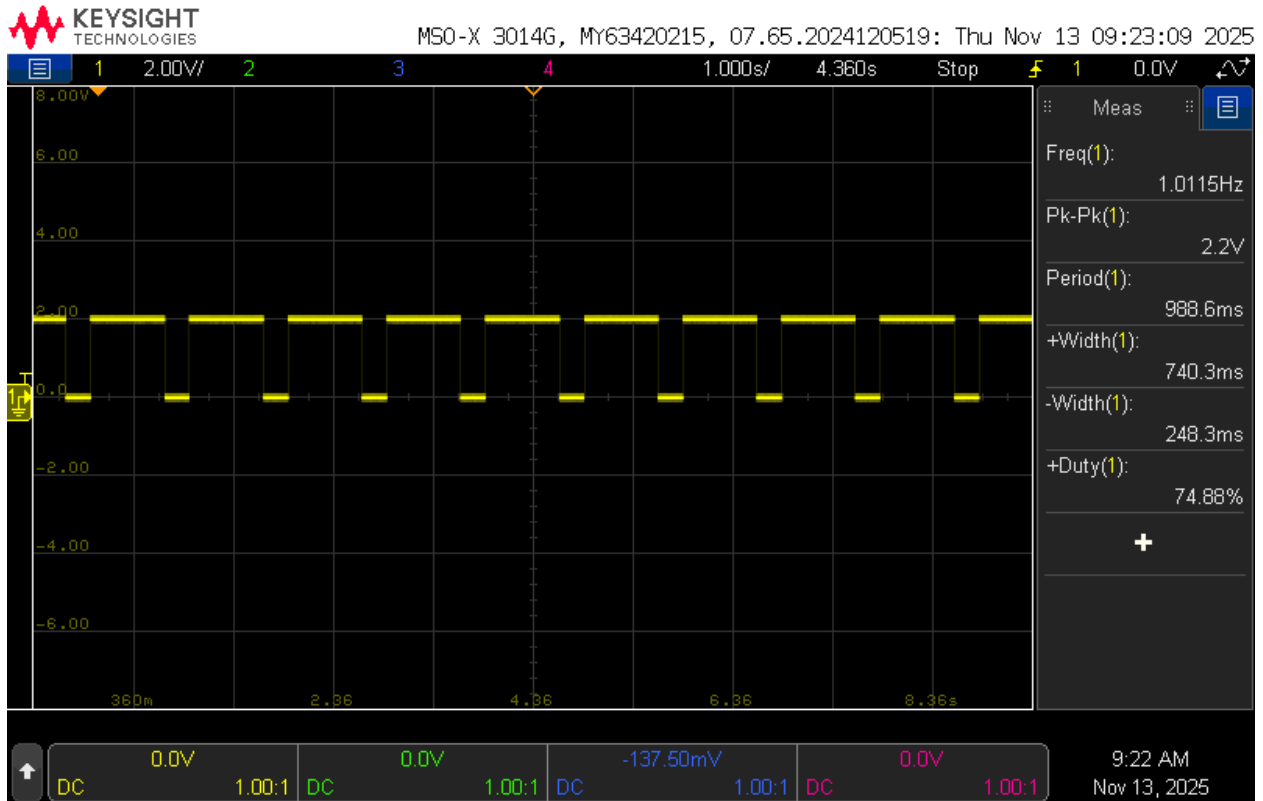


Figure 20: Oscilloscope results for astable simulation in case 2

	T1(s)	T2(s)	T(s)
Theoretical	0.743	0.251	0.994
Simulated	0.750	0.254	1.004
Experimental	0.740	0.248	998.6

Figure 21: Table comparing pulse widths of theoretical, simulated and experimental circuits

Usage with potentiometer

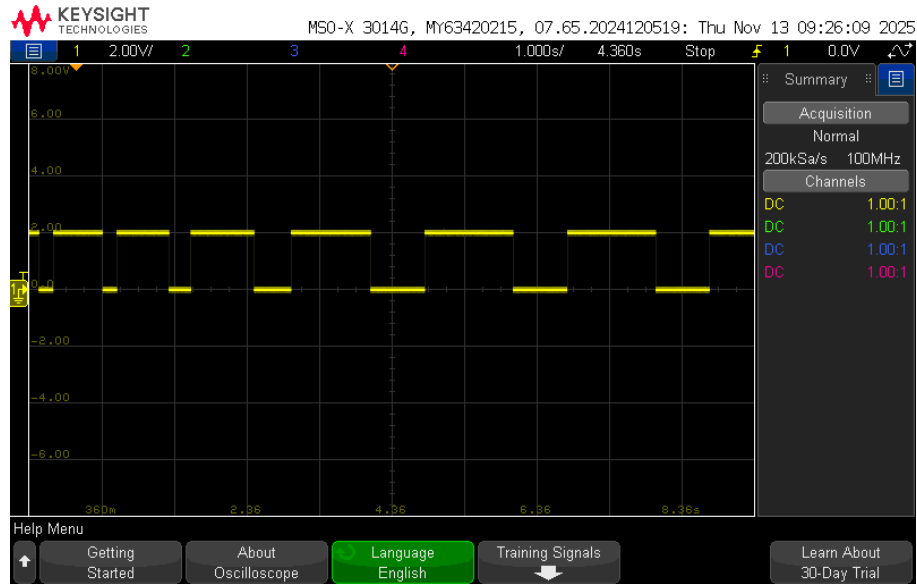


Figure 22: Astable circuit with R2 replaced by potentiometer being adjusted on the fly

555 usage circuit

Because the 4017 is not in LTspice, it is not possible to simulate the circuit. Provided below is a link of two videos showing the dice in action along with oscilloscope outputs showing how quickly the pulse can be created

<https://drive.google.com/file/d/1QQXySA1JY4R2jMCEpBlmyRTQFlwN9uJI/view?usp=sharing>

https://drive.google.com/file/d/1Ycb16NOsYgmZoNIWwjO-6_LWHckebYps/view?usp=sharing

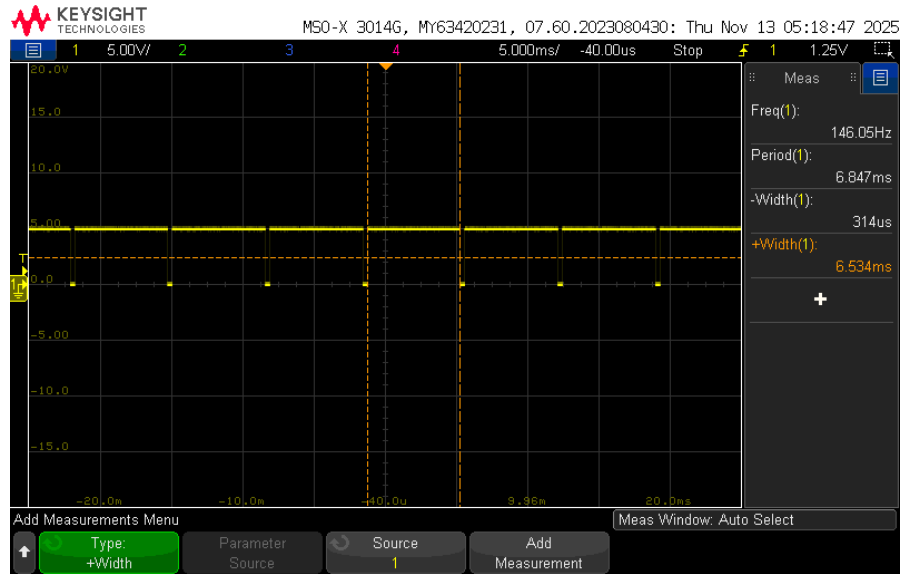


Figure 23: 555 pulse at its highest frequency

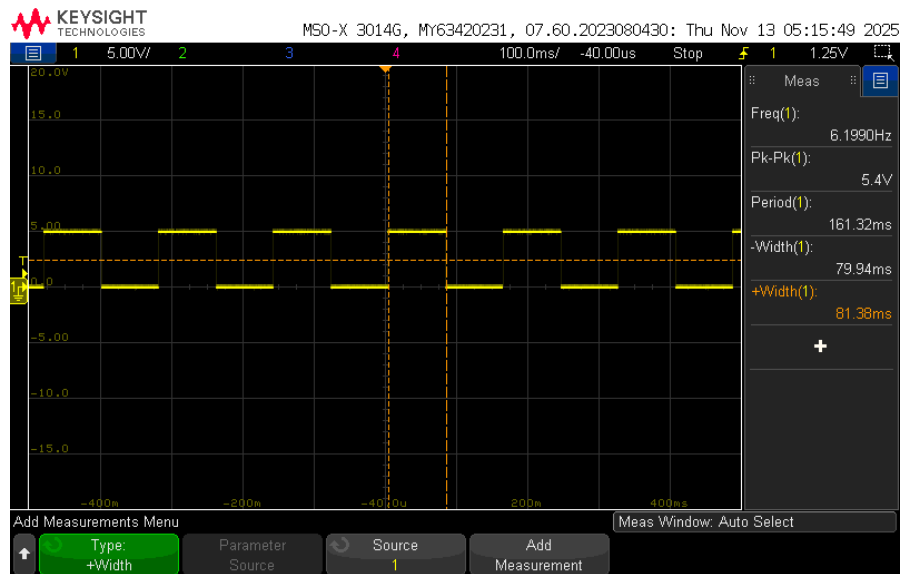


Figure 24: 555 pulse at its lowest frequency

The one problem with this circuit is that it is not a true “random” number generator. Because the numbers are switching quickly enough for a human to not notice, it is not fast enough for a camera/ a computer to not be able to make out the LEDs switching (as seen in the demonstration video). On top of this, the switch does not instantly turn off the connection between the two ICs and at such a high frequency, the last number will always be a couple of numbers ahead of the one at which the button was unpressed. This could, however, be taken as another factor of the randomness of the circuit.

Conclusion

This project successfully calculated and experimentally verified the operation of the 555 timer in its monostable and astable modes. The monostable experiment confirmed the pulse width formula and the astable experiments demonstrated the ability to control frequency and duty cycle, closely matching theoretical formulas. Finally, the main project successfully integrated the 555 timer as an astable pulse generator to create a functional electronic dice with a CD4017 counter.

Most of the values in the experiment are within the 10% permissible error percentage and can mostly be chalked up to the resistance value not being exact.

Improvements to the circuit could be made by debouncing the switch, and decreasing the resistances and capacitances even further to make the circuit higher frequency, although this would come with its own problems.

References

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